

Shashank Obla

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Computer architecture researcher with comprehensive **cross-stack experience**, from custom digital tapeouts to datacenter-scale stochastic modeling and machine learning. Specialized in the design and optimization of **heterogeneous reconfigurable systems** across **networking, security, and database analytics** workloads.

EDUCATION

Carnegie Mellon University, Pittsburgh	2019 - 2026 (<i>Expected</i>)
Ph.D. Candidate, Electrical and Computer Engineering; GPA: 3.97/4.0 <i>PhD Thesis</i> : Design and Optimization of Input-Dependent Streaming Pipelines on FPGAs <i>Advisor</i> : Prof. James C. Hoe	
Indian Institute of Technology, Bombay	2014 - 2019
B.Tech and M.Tech, Electrical Engineering; GPA: 9.84/10.0 Minor in Biosciences and Bioengineering; GPA: 10.0/10.0 <i>Thesis</i> : Accelerated Circuit Simulation [thesis][paper] <i>Advisor</i> : Prof. Sachin Patkar	

AWARDS

3rd Place Winner of Reconfigurable Computing Challenge at FCCM	May 2026
Indian Institute of Technology Institute Gold Medal	2019
Carnegie Institute of Technology Dean's Fellowship	2019

WORK EXPERIENCE

Intel Corporation	Summer 2020
<i>Programmable Solutions Group</i> / <i>Mentor</i> : Scott Weber	
- Modeled Partial Reconfiguration (PR) flows for next-generation FPGAs using SystemC to achieve sub-ms reconfiguration times utilizing existing hardware resources and to assess the benefits of new architectural decisions - Analyzed the full-stack implications of novel PR features by collaborating with cross-functional design teams	

RESEARCH EXPERIENCE

*preprint available

RapidQ: Queuing Abstraction and Tuning for Streaming Pipelines on FPGAs*	2022 - Present
- Designed the RapidQ abstraction for performance modeling of input-dependent streaming pipelines on FPGAs - Devised a modeling framework to extract the model directly from HLS implementations with >97% accuracy - Engineered a queueing-based RapidQ simulator in SystemC achieving >7x speedup over state-of-the-art Reduced resource utilization by over 40% across synthetic as well as real-world network security and machine learning workloads by developing an automated tuning flow powered by RapidQ	
RapidDetect: Threat Detection via FPGA-Accelerated Log Analysis* [paper][code]	2024 - Present
- Developed a heterogeneous FPGA-CPU pipeline for streaming log monitoring using Sigma rules at over 200Gbps - Implemented a highly-parameterizable string matching engine in HLS, tunable for line rates and log characteristics - Achieved a 100x reduction in threat detection latency and reduced costs by over 10,000x compared to commercial alternatives using a single FPGA server by integrating the FPGA string-matching filter with Hyperscan	
Group-by Aggregation Accelerator with NoC-enabled Partitioning	2025 - Present
- Architected a streaming group-by aggregation pipeline capable of processing over 64GBps on a single FPGA - Designed and open-sourced a parameterizable Network-on-Chip in RTL operating at over 500MHz, which has been used for hash-based tuple partitioning and also in the development of the NoC-based OpenFPGA by other groups - Reduced bandwidth by over 2x for skewed workloads with hot keys by introducing a fully-associative pre-aggregator	

SELECTED PUBLICATIONS

Conference Proceedings

- [Shashank Obla](#), Bin Li, James C. Hoe. “RapidQ: Queueing-based Performance Modeling Framework for Rapid Simulation and Automated Tuning of Input-Dependent Streaming FPGA Pipelines.” *Under Review*.
- Shashank Obla, Tommy Tracy II, Matthew Beck, James C. Hoe, Kevin Skadron, Wajih Ul Hassan. “RapidScan: High-Throughput Parameterized HLS-based Streaming String Matching Library for FPGAs.” *Reconfigurable Computing Challenge, FCCM*, May 2026.
- Zhipeng Zhao, Joseph Melber, Siddharth Sahay, [Shashank Obla](#), Eriko Nurvitadhi, James C. Hoe. “Exploiting the Common Case When Accelerating Input-Dependent Stream Processing by FPGA.” *IEEE Transactions on Computers*, May 2023.

Posters

- Shashank Obla, Bin Li, James C. Hoe. “Lightweight Queueing Abstraction for Rapid Simulation and Automated Tuning of Input-Dependent Streaming Pipelines on FPGAs.” *The 34th IEEE International Symposium on Field-Programmable Custom Computing Machines*, May 2026.

TEACHING EXPERIENCE

Evidence Based Teaching in STEM

2024

Course offered by Eberly Center for Teaching Excellence & Educational Innovation at CMU

- Developed a new course including learning objectives-driven syllabus design and inquiry-based learning assignments
- Refined teaching skills through practice teaching sessions and critically reviewed Scholarship in Teaching and Learning

Performance Modeling Tools for Computer Systems Researchers

2022 (volunteer)

Graduate Teaching Assistant | Prof. Mor-Harchol Balter

- Mentored 20+ PhD students in **applying queuing theory** to their systems research through one-on-one meetings
- Designed homework assignments to enable students to develop and verify insights through queuing system simulations

Reconfigurable Logic: Technology, Architecture, and Applications

2020, 2021, 2022 (volunteer)

Graduate Teaching Assistant | Prof. James C. Hoe

- Developed a DFX-capable Vitis platform for Ultra96v2 enabling students to gain first-hand experience in using role-and-shell partial reconfiguration in the lab exercises and in their course projects
- Redesigned the recitation syllabus and conducted recitations introducing system design concepts for FPGAs

OTHER PROJECTS

Accelerating Stencil Computation – Multigrid V-Cycle Poisson Solver [\[pdf\]](#)

2022

- Achieved **>140x speedup** (over 50% of arithmetic peak) using **SIMD intrinsics** compared to [Proto](#) from LBNL

Enabling Design Tradeoffs at Runtime with Dynamic Composition | *ECE Qualifiers* [\[pdf\]](#)

2021

- Implemented a **hierarchical PR** architecture to dynamically tune modules to changing constraints and opportunities showcasing **>3x improvement** in performance for a Breadth-First Search accelerator over static provisioning

FPGA Tape-Out in TSMC 28nm | *Course Project funded by Apple*

2020

- Designed, taped out and post-silicon verified a **custom FPGA fabric** supported by the VTR software toolchain

SERVICE AND LEADERSHIP

Student Council for Faculty Hiring | CMU ECE

2023 - 2024

- Chaired the first such student council at ECE, **managing** a group of **25 students** in conducting faculty **interviews**

Committees

- FCCM Artifact Evaluation Committee

2026

TECHNICAL SKILLS

Programming	System Verilog, Vitis/Altera HLS, SystemC, C++, Assembly/Intrinsics, Python Antigravity
Design Tools	Quartus, Vivado, VTR Toolchain, Platform Designer, Cadence - Genus/Innovus/Voltus/Virtuoso